

Name

Class

Date

OOP - 13

Total questions: 10

Worksheet time: 7mins

1.

```
class A {
public:
    A() {cout << "A ";}
    ~A() {cout << "~A ";}
};

class B : public A {
public:
    B() {
        cout << "B ";
        throw -1;
    }
    ~B() {cout << "~B ";}
};

int main() {
    try {
        B b;
    }
    catch (...) {
        cout << "error caught ";
    }
}
```

a) A
B
~A
error caught

c) A
B

b) A
B
error caught

d) A
B
~B
~A
error caught

2.

```
class A {
    int a;
public:
    A(int _a) : a(_a) {cout << "A ";}
};

class B {
    int b;
public:
    B(int _b) : b(_b) { cout << "B ";}
};

class C : public A, public B {
public:
    C(int _a, int _b) : B(_b), A(_a) { cout << "C ";}
};

int main() {
    C c(1, 2);
}
```

a) C

B

A

c) A

B

C

b) B

A

C

d) C

3.

```
class A {
public: A() { cout << "A ";} };

class B : virtual public A {
public: B() { cout << "B ";} };

class C : virtual public A {
public: C() { cout << "C ";} };

class D : public C, public B {
public: D() { cout << "D ";}
};

int main() {
    D d;
}
```

a) C

B

D

c) A

C

B

D

b) C

B

A

D

d) A

B

C

D

4.

```
template<typename T>
class Calculator {
public:
    static T difference(const T& obj1, const T& obj2) {
        return obj1 - obj2;
    }
};

template<>
class Calculator<char> {
public:
    static char difference(const char& obj1, const char& obj2) {
        return '0';
    }
};

int main() {
    cout << Calculator<int>::difference(2, 1) << " ";
    cout << Calculator<char>::difference('b', 'a');
}
```

a) 1 1

b) 1 0

c) 1 a

d) Няма да се компилира!

5.

```
class A {
public:
    A(int a) {cout << "A ";}
};

class B : public A {
public:
    using A::A;
};

int main() {
    B b(2);
}
```

a) A

b) нищо

c) Няма да се компилира!

6.

```
int main() {
    unique_ptr<int> ptr = make_unique<int>(1);
    cout << *ptr;
}
```

a) 1

b) адрес

c) нищо

d) Няма да се компилира!

7.

```
class A {
public:
    A() {cout << "A() ";}
    A(const A& obj) {cout << "copyA() ";}
    virtual void f() {cout << "A ";}
    virtual ~A() {}
};

class B : public A {
public:
    B() {cout << "B() ";}
    B(const B& obj) {cout << "copyB() ";}
    void f() override {cout << "B ";}
    ~B() {}
};

void g(A obj) {
    obj.f();
}

int main() {
    unique_ptr<A> ptr = make_unique<B>();
    g(*ptr);
}
```

a) A()

B()

B

c) A()

B()

copyA()

A

b) A()

B()

copyB()

B

d) Няма да се компилира!

8.

```
class A {
public:
    A() {cout << "A() ";}
    A(const A& obj) {cout << "copy A() ";}
    virtual void f() {cout << "A ";}
    virtual ~A() {}
};

class B : public A {
public:
    B() {cout << "B() ";}
    B(const B& obj) {cout << "copy B() ";}
    void f() override {cout << "B ";}
    ~B() {}
};

void g(B obj) {
    obj.f();
}

int main() {
    A obj;
    g(obj);
}
```

a) A()

copyB()

A

c) A()

copyB()

B

b) A()

copyA()

A

d) Няма да се компилира!

9.

```
class A {
    int a = 0;
public:
    A(int _a) : a(_a) {}
    int getA() {return a;}
};

void f(const A& obj) {
    cout << obj.getA();
}

int main() {
    A obj(67);
    f(obj);
}
```

a) 67

b) 0

c) Няма да се компилира!

10.

```
int main() {
    shared_ptr<int> ptr1 = make_shared<int>(1);
    {
        cout << ptr1.use_count() << " ";
        auto ptr2 = ptr1;
        cout << ptr2.use_count() << " ";
    }
    cout << ptr1.use_count() << " ";
}
```

a) 1 2 1

b) 1 1 1

c) 1 1 3

d) Няма да се компилира!

Answer Keys

- | | | |
|---------------------------|-----------------------------|-----------------------------|
| 1. a) A B ~A error caught | 2. c) A B C | 3. c) A C B D |
| 4. b) 1 0 | 5. a) A | 6. a) 1 |
| 7. c) A() B() copyA() A | 8. d) Няма да се компилира! | 9. c) Няма да се компилира! |
| 10. a) 1 2 1 | | |

